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(a)(i)	A photon has an energy which is dependent on its frequency ($E = hf$), which is lower than the work function, hence not sufficient emit the electrons.
	For waves, the energy is dependent on the amplitude/intensity of the wave.
(a)(ii)	A single photon is absorbed by a single electron and the interaction is almost instantaneous.
	For waves, it will take time for the photon to absorb and accumulate sufficient energy to be ejected.
(b)(i)1.	$E = \frac{hc}{\lambda}$ $= \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{(680 \times 10^{-9})}$
	$E = 2.9 \times 10^{-19} \text{ J}$
(b)(i)2.	$I = \frac{P}{A} = \frac{\left(\frac{N}{t}\right)\left(\frac{hc}{\lambda}\right)}{\pi\left(\frac{d}{2}\right)^2}$ $\frac{N}{t} = \frac{(3.2 \times 10^3)\pi\left(\frac{1.5 \times 10^{-3}}{2}\right)^2}{2.9 \times 10^{-19}}$
	$\frac{N}{t} = 1.9 \times 10^{16} \text{ s}^{-1}$
(b)(ii)	$p_i = \frac{h}{\lambda} \cos 30^\circ$ $= \frac{(6.63 \times 10^{-34})}{(680 \times 10^{-9})} \cos 30^\circ$ $= 8.4 \times 10^{-28} \text{ kg m s}^{-1}$
	$F_{\text{on one photon}} = \frac{p_f - p_i}{t} = \frac{(-8.4 \times 10^{-28}) - (8.4 \times 10^{-28})}{t}$
	$F_{\text{on surface by one photon}} = -\left(\frac{p_f - p_i}{t}\right) = \frac{1.7 \times 10^{-27}}{t}$
	$\text{total force on surface by photons} = -N\left(\frac{p_f - p_i}{t}\right)$ $= (1.9 \times 10^{16})(1.7 \times 10^{-27})$ $= 3.3 \times 10^{-11} \text{ N}$
(b)(iii)	The reflected photons will have a <u>smaller change in momentum</u> , hence the force will be smaller.

Anglo-Chinese Junior College
H2 (9749) Physics

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